

FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in Cellular Reprogramming



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ABSTRACT

Cellular reprogramming with delivered transcription factors underpins regenerative medicine, but the endogenous regulators acting alongside them remain only partly known. Existing models infer such regulators from network topology or expression dynamics, without learning from the factors known to drive the system. Here we present **FateDriver**, which takes the delivered factors as its positive labels, learns how they align with the source-to-target direction, and returns the unlabeled genes that align the same way as candidates for a shared role in the transition. Across three systems it recovers externally curated regulators better than an expression-matched baseline, separating transcriptional similarity from regulator recovery. A score belongs to a transition: FateDriver predicts oppositely directed programs for two fibroblast states of one heterogeneous culture, and different programs for the partial and iPSC endpoints. Transferred to another dataset, it ranks candidate origins for a population partially reprogrammed *in vivo*, placing the intestinal stem cell first.

RESULTS

I. FateDriver : a dual-view graph encoder which scores genes against a transition

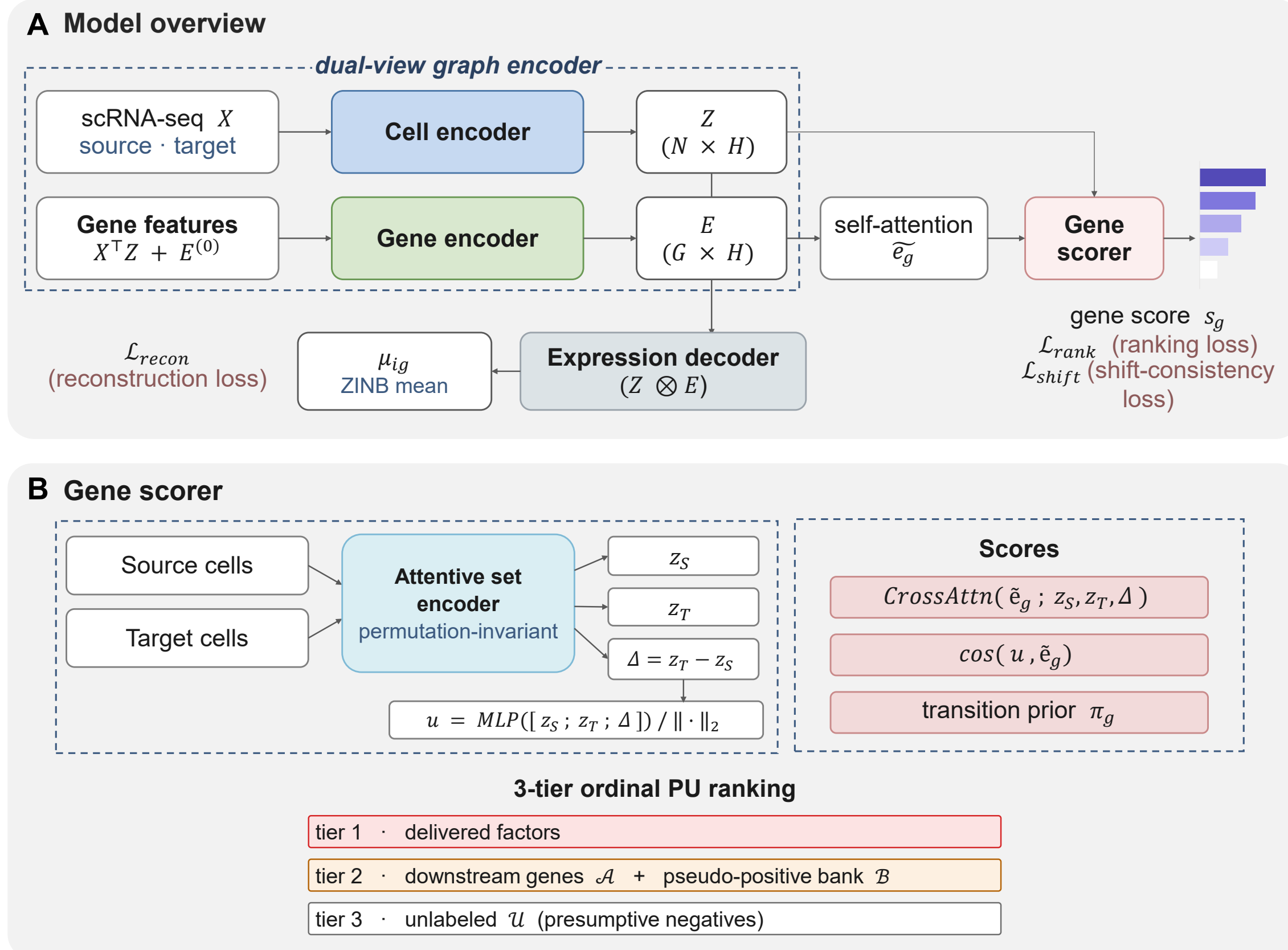


Figure 1. The FateDriver framework. (A) Source and target transcriptomes with a pseudotime ordering enter a dual-view graph encoder — a GCN over a pseudotime-directed cell graph and a GCN over the NicheNet prior gene graph — which places cells and genes in one latent space; a ZINB decoder keeps both embeddings anchored to the data without entering the score. (B) Gene scorer: a permutation-invariant set encoder pools each population, three additive terms score every gene against the transition, and a three-tier positive–unlabeled objective ranks the delivered factors above their downstream targets above the unlabeled remainder.

II. FateDriver recovers curated regulators without leading on co-expression with the delivered factors

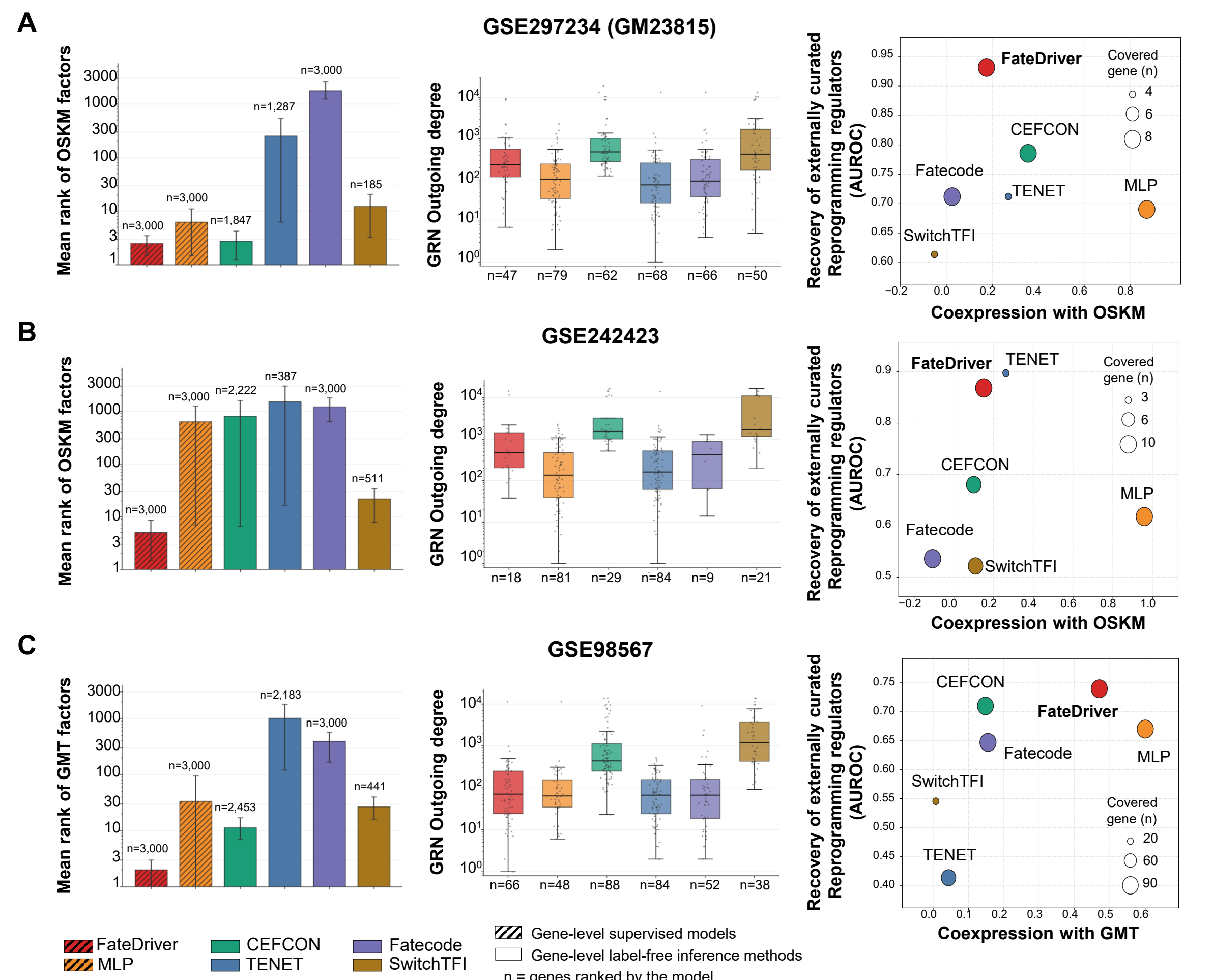


Figure 2. Benchmark against four label-free methods and an expression-matched baseline. Rows are datasets. Hatched bars are the two gene-level supervised models. Left, mean rank of the delivered factors (lower is better, log axis). Middle, NicheNet outgoing degree of each method's top-100 genes. Right, co-expression with the delivered-factor module (x) against recovery of an externally curated regulator set (y , AUROC); marker size grows with the number of curated positives the AUROC rests on.

III. FateDriver ranks genes differently for two fibroblast source states sharing one iPSC target

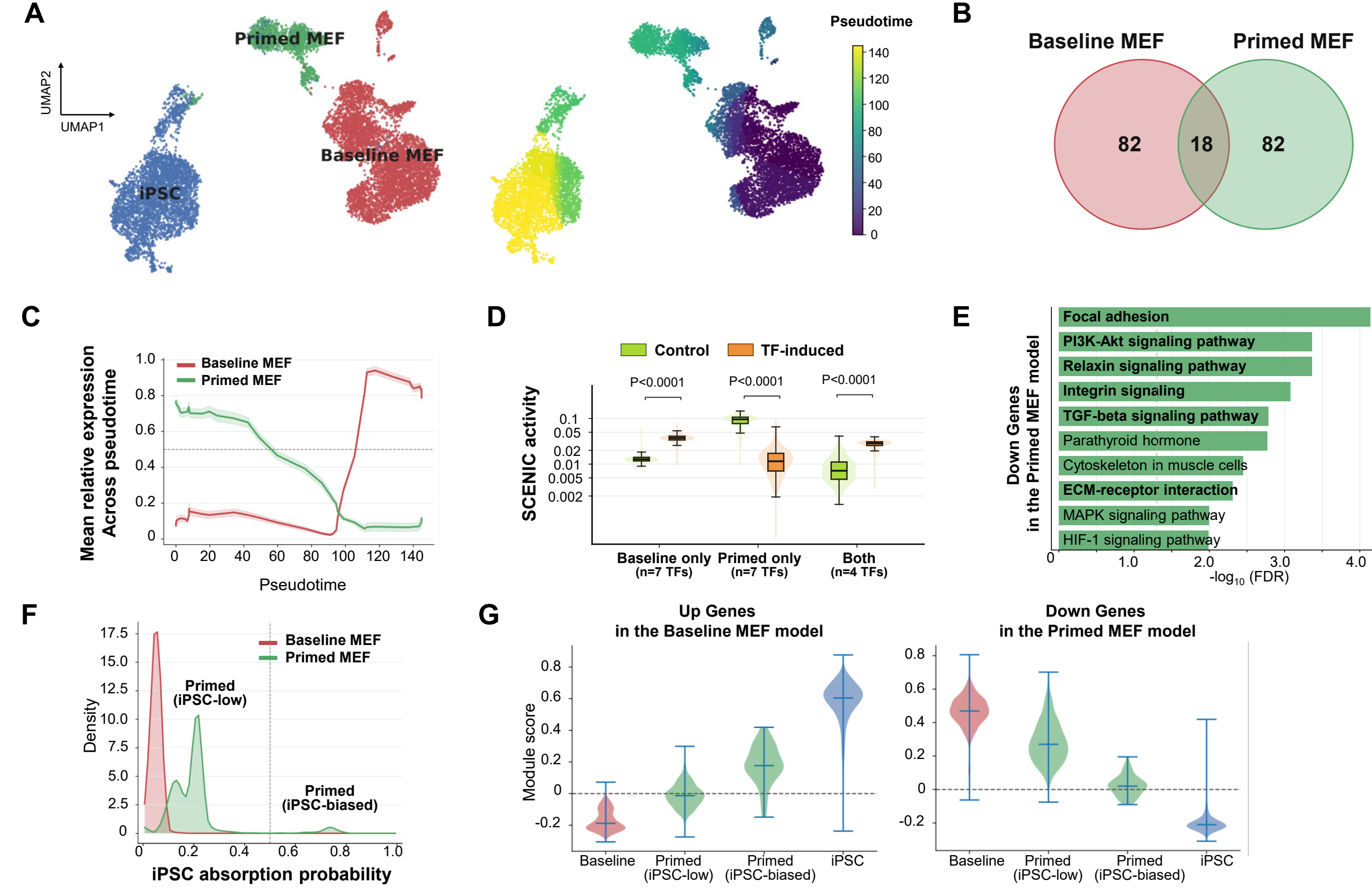


Figure 3. Source-state conditioning of the ranking (GSE181451, MEF → iPSC). (A) UMAP by cell annotation and by pseudotime. (B) The two top-100 cells share only 18 genes — the delivered factors and the senescence barrier. (C) Baseline-specific genes switch on late along pseudotime; Primed-specific genes switch off early. (D) pySCENIC regulon activity in control (MEF) and TF-induced (iPSC) cells, scored from each factor's targets rather than its own transcript. (E) KEGG terms of the switch-off Primed-specific genes. (F) CellRank iPSC absorption probability, bimodal in Primed MEF. (G) Each signature keeps its direction across four groups ordered by iPSC propensity, including the two Primed fractions neither model saw.

IV. FateDriver separates partial from complete reprogramming, and interferon activity marks cell-to-cell variation in outcome

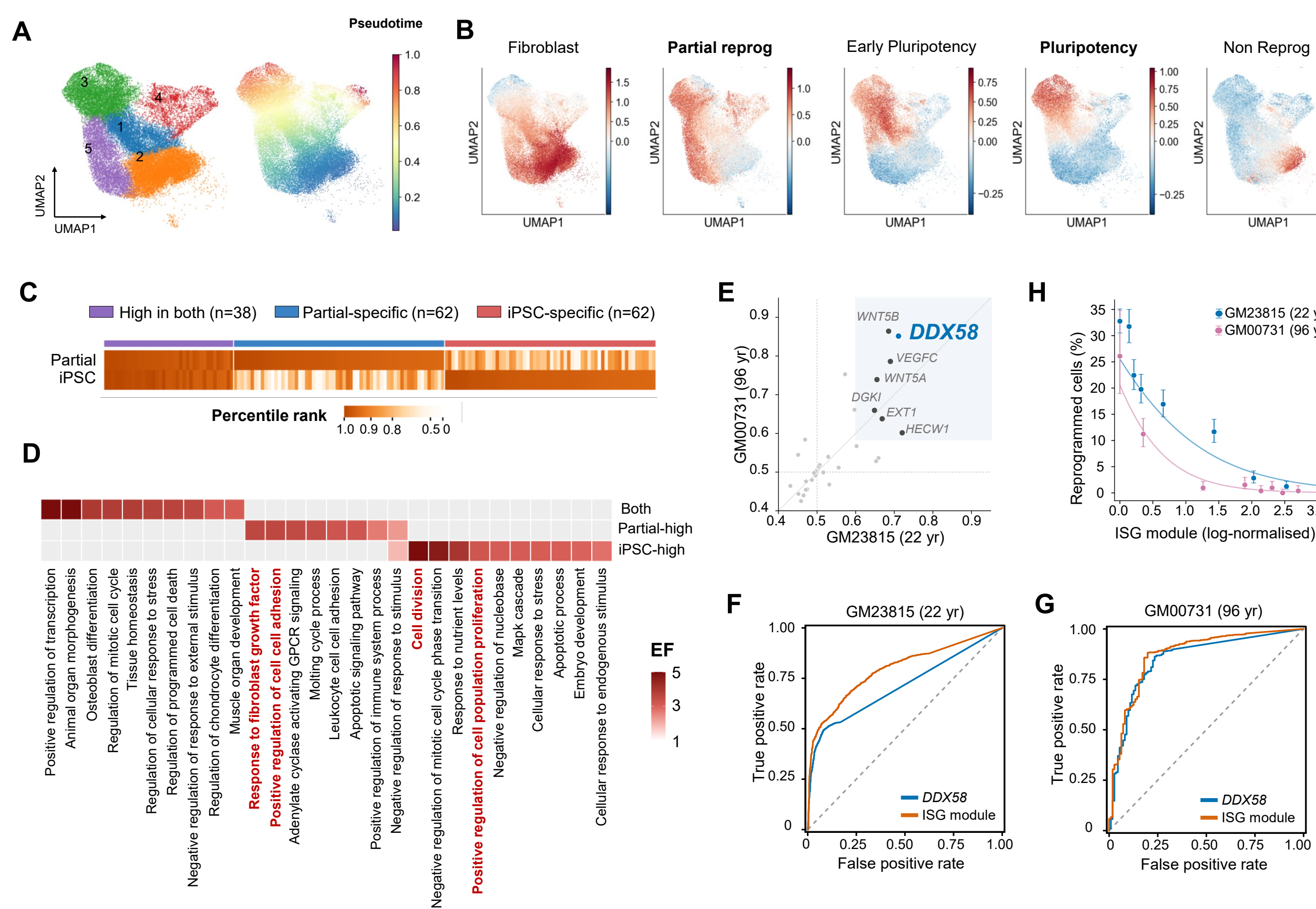


Figure 4. Two endpoints reached from one fibroblast culture (GSE297234, GM23815). (A) Cell clusters and pseudotime over the four integrated timepoints (D0, D3, D7, D10). (B) Module scores distinguishing the reprogramming states. (C) Rank percentiles along the two trajectories give a shared Both-high core ($n = 38$) and two condition-specific sets of 62 genes. (D) GO:BP terms: cell-cell adhesion for the partial set, cell division for the iPSC set. (E) Screen of the Partial-specific genes against day-10 outcome; DDX58 ranks first. (F, G) ROC for DDX58 and a four-gene ISG (interferon-stimulated gene) module (ISG15, IFIT1, IFIT3, IRF7). (H) The reprogrammed fraction falls with ISG activity in both donors.

V. FateDriver ranks CBC highest among candidate origins of the partially reprogrammed aVEC population

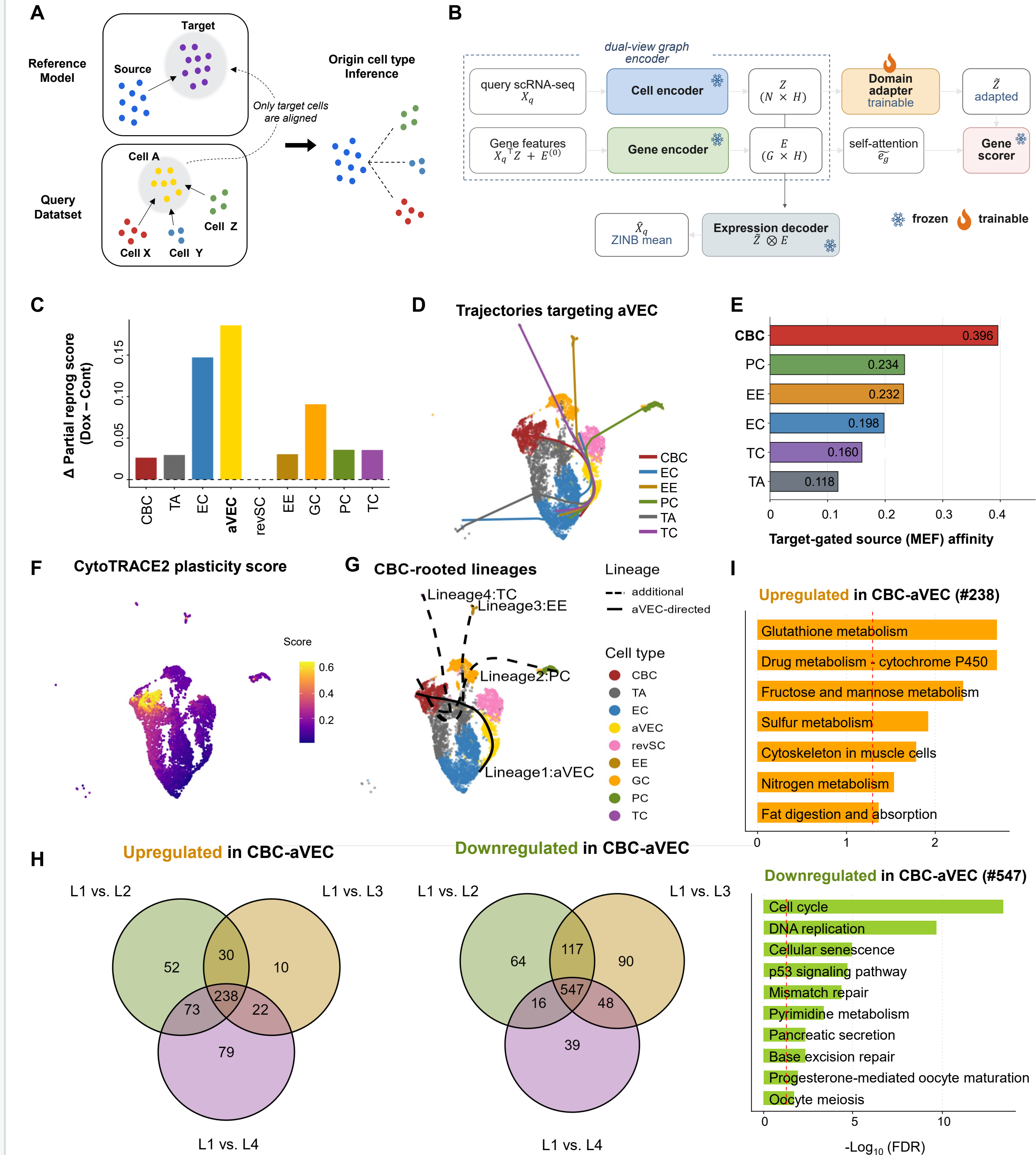


Figure 5. Transfer of the Baseline MEF model to an *in vivo* partially reprogrammed intestinal population (GSE237173). (A) The query target cells alone are aligned to the reference iPSC pole; each candidate epithelial type is then scored at the reference MEF pole. (B) Domain-adaptation architecture used for the transfer (C) Change in partial-reprogramming module score on OSKM induction (Dox – Cont) per cell type; aVEC responds most. (D) Six trajectories with aVEC as the target, one from each candidate source. (E) Target-gated source affinity ranks CBC first (0.396, about 1.7× the next candidate). (F) CytoTRACE2, a criterion the transfer does not use, gives CBC the highest developmental plasticity. (G) Four lineages resolved with CBC as the source, the CBC → aVEC lineage recovered among them. (H) Genes consistently up- ($n = 238$) or down-regulated ($n = 547$) in that lineage against the other three. (I) KEGG enrichment of the two sets: metabolic programs up, cell cycle and DNA replication down.

SUMMARY

- What it does:** From a source population, a target population, and the few factors a protocol delivers (OSKM, GMT), FateDriver learns what those factors do along the transition and ranks every other gene that behaves the same way.
- Not limited by prior networks:** Existing methods rank a gene only if a prior GRN or TF annotation contains it (185–2,453 of 3,000 genes); FateDriver ranks all 3,000 from a space learned from the data itself.
- Recovers real regulators, not look-alikes:** It recovers curated reprogramming enhancers and barriers best or second-best in three datasets, while a baseline picking genes co-expressed with OSKM/GMT does not.
- A score belongs to a transition, not to a gene:** Two fibroblast states of one culture, scored against the same iPSC target, shared only 18 of their top 100 genes and moved in opposite directions along pseudotime; partial and complete endpoints gave two distinct programs (cell–cell adhesion vs cell division).
- Transferable to a new system without retraining:** A trained model is carried onto another dataset by fitting one small adapter, leaving the learned axes intact.